

Serkan Shentyurk

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[LinkedIn](#)
[GitHub](#)

PROFILE

MSc in Statistics & Data Science (*cum laude*) with 3+ years of experience developing statistical and machine-learning models for experimental data. Strong Python background across predictive modelling, simulation-based inference, state-space models, and reproducible data workflows. Experienced with scikit-learn, XGBoost, Git, Docker, and HPC, with working knowledge of SQL.

EXPERIENCE

PhD Researcher — Akrami Lab, University College London

2024 – present

Developing statistical and machine-learning methods to model trial-level behaviour and learning from experimental data.

- Replaced a data-hungry grid-search estimator with engineered summary statistics and simulation-based inference (**sbi**), reducing the data required per estimate to **a single session (~300 trials)** and extending the method to learners where the original approach was unreliable.
- Developed **behav_utils**, a reusable Python package for behavioural-data analysis, and built a reproducible HPC pipeline using SLURM array jobs (**sound_categorisation**: 396-test suite).
- Applied state-space models to longitudinal behavioural data to identify **recurring learning stages**, then used a disentangled RNN (**disRNN**) to infer latent decision variables and test model-derived hypotheses.
- Designed and analysed a causal intervention to test its effect on behaviour and inferred strategy.

Research Assistant — Gonçalves Lab, VIB AI

2023 – 2024

Computational modelling of excitatory–inhibitory neural networks against experimental data to study changes in network behaviour under perturbation.

- Built and simulated excitatory–inhibitory network models and fitted model parameters against observations.
- Combined machine learning, parameter search, and statistical hypothesis testing to evaluate competing explanations for changes in network behaviour. [\[repo\]](#)

PROJECTS

Bank Marketing — Imbalanced Classification and Calibration

[\[repo\]](#)

Python · scikit-learn · XGBoost · Docker

Ranked-call-list model on the UCI Bank Marketing dataset; 41,188 records with ~11% positive class.

- Ranked clients by subscription probability (logistic regression vs XGBoost, stratified cross-validation, PR-AUC): **calling the top 10% converts at 50% — $4.5\times$ the 11% base rate.**
- Calibrated the probabilities and set the decision threshold from a call budget rather than 0.5, turning scores into a subscription-capacity forecast (**~206 calls to land 150 subscriptions**).
- Containerised the analysis with Docker for a reproducible modelling pipeline.

Eel-Fisher Survey — Statistical Testing and Predictive Modelling

[\[repo\]](#)

Python · scikit-learn · statsmodels

Commissioned statistical analysis for a conservation organisation; $n = 51$ across two independent questionnaires.

- Harmonised two questionnaires into a 24-field typed dataset and performed type-aware two-sample tests with multiple-comparison correction, identifying **eight fields with significant between-group differences**.
- Built regularised logistic-regression models with leakage-safe preprocessing and evaluated limitations caused by small n , near-complete separation, and single-split evaluation.

SKILLS

Programming: Python · SQL · R · Git

Statistics & modelling: GLMs · time-series · state-space models · simulation-based inference · hypothesis testing · probability calibration

Machine learning: scikit-learn · XGBoost · classification · clustering · dimensionality reduction · cross-validation

Data & visualisation: pandas · NumPy · statsmodels · Plotly · matplotlib

Engineering: Docker · SLURM / HPC

EDUCATION

PhD, Computational and Systems Neuroscience — UCL

2024 – present

Wellcome PhD Studentship

MSc, Statistics and Data Science — KU Leuven

2022 – 2024

Cum laude · Peter Rousseeuw Scholarship

BSc, Molecular Biology and Genetics — Bilkent University

2018 – 2022

Summa cum laude